Amino Acids

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Amino Acids

Protein (Amino Acid)

H_2N-CH_2-COOH	α amino acid
$H_2N-CH_2-CH_2-COOH$	β amino acid
$H_2N-CH_2-CH_2-CH_2-COOH$	γ amino acid

All amino acid in human body are constituently alpha-amino acid.
Compound which contain amine as well as
carboxylic acid are known as Amino Acid.

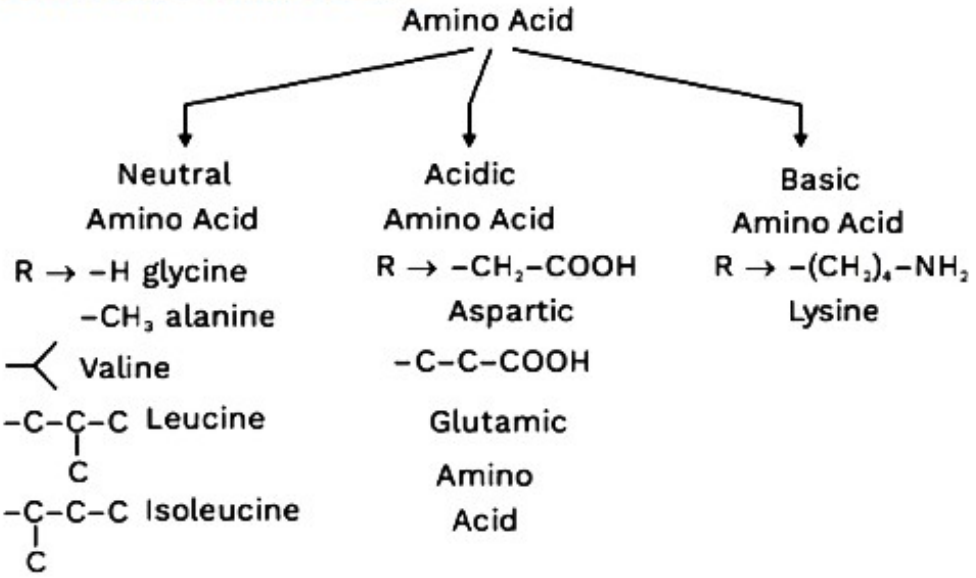


(Zwitter ion/ Dipole ion/Inner salt)

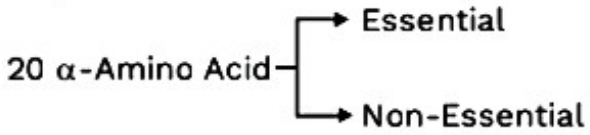
Points to remember

* Solubility in H_2O is high
* Crystalline solid
* High meting point

Classification of Amino Acid:



Glycine is only amino acid which is optically inactive because it has no chiral centre.
Cysteine is the only L-amino acid which has R configuration rest all L-amino acid have configuration.
Total 20 α-amino acids are required in the formation of protein in human body.



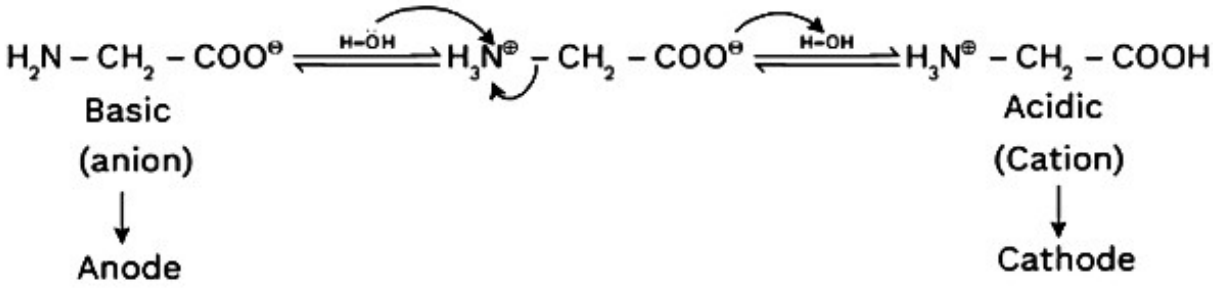
Essential Amino Acid :

Which human body cannot prepare on its own and we require from outside 10 essential amino acids.

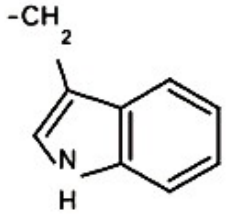
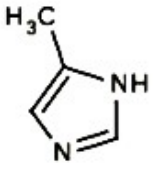
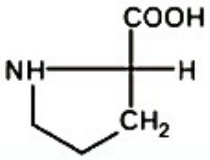
Amino acids which cannot be synthesised in our body and we take these amino acid through diet are essential amino acids.

Amino Acid in water :

Exist in three forms, that's why it can be migrated towards cathode and anode.



Name of the amino acids		Characteristic feature of side chain, R	Three letter symbol	One letter code
1.	Glycine	H	Gly	G
2.	Alanine	$-CH_3$	Ala	A
3.	Valine	$(H_3C)_2\,CH-$	Val	V
4.	Leucine	$(H_3C)_2\,CH-CH_2-$	Leu	L
5.	Isoleucine	$H_3C-CH_2-CH(CH_3)-$	Ile	I
6.	Arginine*	$HN=C(NH_2)-NH-(CH_2)_3-$	Arg	R
7.	Lysine*	$H_2N-(CH_2)_4-$	Lys	K
8.	Glutamic acid	$HOOC-CH_2-CH_2-$	Glu	E
9.	Aspartic acid	$HOOC-CH_2-$	Asp	D
10.	Glutamine	$H_2N-C(=O)-CH_2-CH_2-$	Gln	Q
11.	Asparagine	$H_2N-C(=O)-CH_2-$	Asn	N
12.	Threonine*	$H_3C-CHOH-$	Thr	T
13.	Serine	$HO-CH_2-$	Ser	S
14.	Cysteine	$HS-CH_2-$	Cys	C
15.	Methionine*	$H_3C-S-CH_2-CH_2-$	Met	M

16.	Phenylalanine*	$C_6H_5 - CH_2 -$	Phe	F
17.	Tyrosine	$(p)HO - C_6H_4 - CH_2 -$	Tyr	Y
18.	Tryptophane*		Trp	W
19.	Histidine		His	H
20.	Proline		Pro	P

- Essential; amino acid, a = entire structure

Isoelectric point :

pH at which particular amino acid is not influenced by electric field is known as isoelectric point.

At isoelectric point [Cation] = [Anion]

At isoelectric point net charge on amino acid is zero.

At isoelectric point maximum concentration of zwitter/dipole ion.

$$H_3N^{\oplus} - CH_2 - COOH \underset{K_1}{\overset{-H^{\oplus}}{\rightleftharpoons}} H_3N^{\oplus} - CH_2 - COO^{\ominus} + H^{\oplus}$$

(Cation) (DI)

$$H_3N^{+}-CH_2-COO^{\ominus} \underset{}{\overset{K_2}{\rightleftharpoons}} H_2N-CH_2-COO^{\ominus} + H^{+}$$

(DI) (Anion)

$$K_1 = \frac{[DI][H^{\oplus}]}{[Cation]}$$

$$K_2 = \frac{[Anion][H^{\oplus}]}{[DI]}$$

At isoelectric point : [Cation] = [Anion]

$$\frac{[DI][H^{\oplus}]}{K_1} = \frac{K_2[DI]}{[H^{\oplus}]}$$

$$[H^+]^2 = K_1K_2$$

$$[H^+] = \sqrt{K_1K_2}$$

$$\text{Log}\,[H^+] = \text{Log}\sqrt{K_1K_2}$$

$$-pH = \frac{1}{2}(-pK_1 - pK_2)$$

$$\boxed{pH = \frac{pK_{a_1} - pK_{a_2}}{2}}$$

↓

of isoelectric point

$$pK_a = \text{Log}\,K_a.$$

Calculation of Isoelectric point :

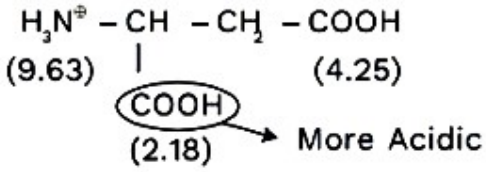
Q3 $H_3N^{\oplus} - CH_2 - COO^{\ominus}$

Sol. $H_3N^{\oplus} - CH_2 - COO^{\ominus}$

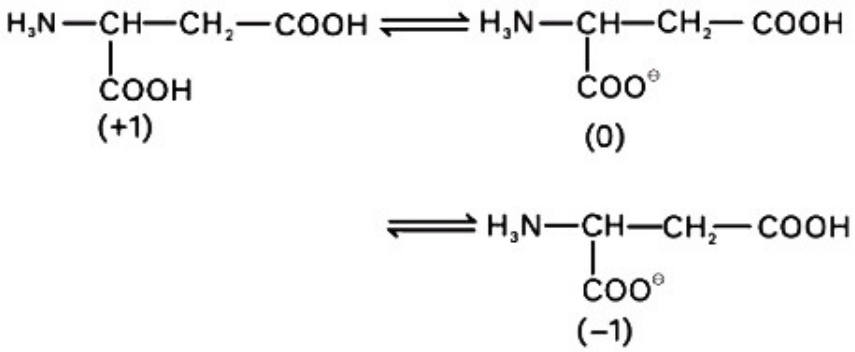
$(pK_a = 9)$ $(pK_a = 3)$

$$PI = \frac{9+3}{2} = 6$$

Q4

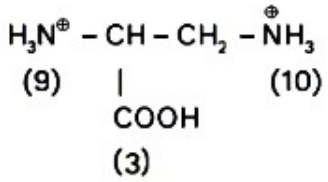


Sol.

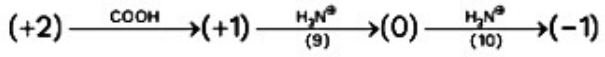


$$PI = \frac{2.18 + 4.25}{2} = \frac{6.43}{2} = 3.215$$

Q5

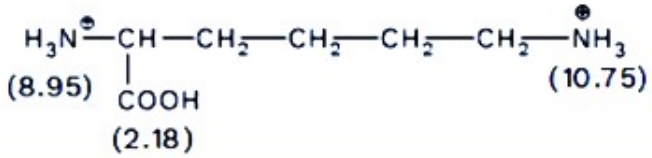


Sol.

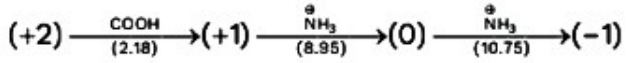


$$PI = \frac{19}{2} = 9.15$$

Q6



Sol.

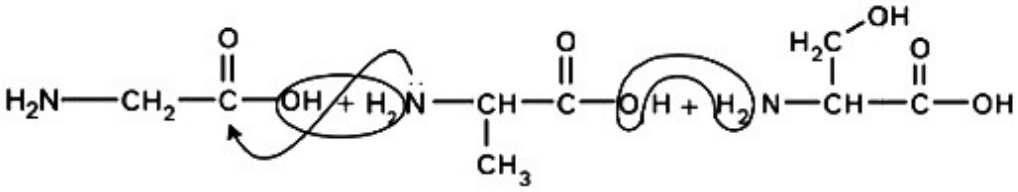


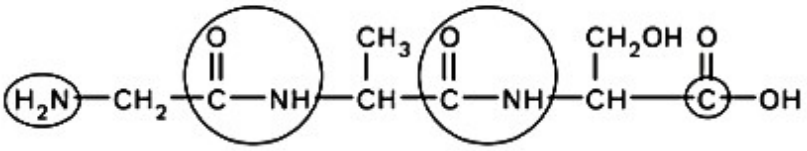
$$P_I = \frac{2.95 + 10.75}{2} = 9.85$$

Types of Protein

- Primary structure (Polypeptide)
- Polypeptide means amide bond of α-amino acids
- Protein is written from N terminal to C-terminal.

  Amino Acid ⟶ Glycine + Alanine + Serine





N – terminal to C – terminal

N-amino acid

(n-1) peptide linkage (Amide bond)

(n-1) H_2O

3 Amino Acid

(3) tripeptide

Calculation of Polypeptide

Possible Polypeptides = X^n → di/tri/tetra etc.

X^n → Number of amino acid

1. Glycine + Alanine

Dipetide $\Rightarrow 2^2 = 4$

GG

AA

GA

AG

Tripeptide $\Rightarrow 2^3 = 8$

GGG

AAA

GAA

AGA

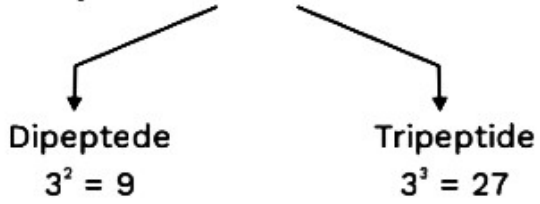
AGG

GGA

GAG

AAG

2. Glycine + Alanine + Serine



3. 10 amino acid octapeptids → 10^8

4. dipeptide $\xrightarrow{H_2O^{\oplus}}$ glycine + Alanine

GA
AG
} 2

5. Tripetide $\xrightarrow{H_2O^{\oplus}}$ Serine + Alanine + Threonine

↳ 6 ways

SAT
STA
ATS
AST
TAS
TSA
} 6

	Fibrous	Globular
1	They are linear structures held together by H-bonding and sulphur linkages	These are coiled shaped structures
2	They are water insoluble	They are water soluble
3	Eg. Keratin (present in hair) myosin (present in muscles)	E.g. Insulin and Albumin

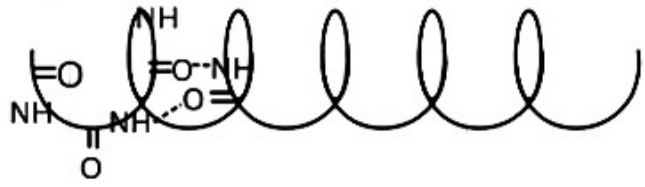
Types of Globular protein :

(i) Primary structure → Fibrous
Already taught

(ii) Secondary Structure

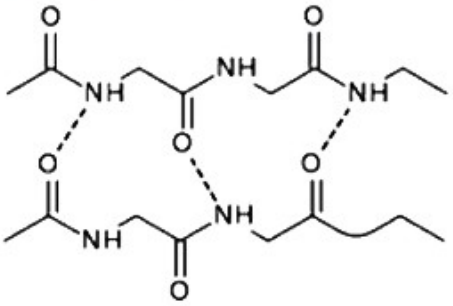
Secondary Structure → α-Helix
Secondary Structure → β-Pleated sheet

α-Helix



Intramolecular H-Bonding

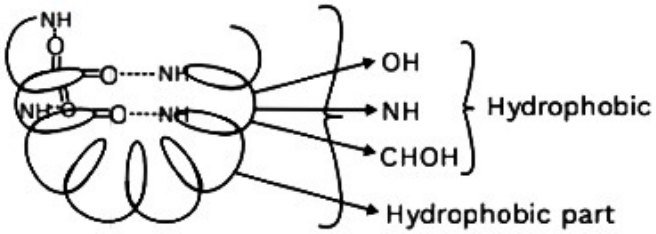
β-Pleated sheet

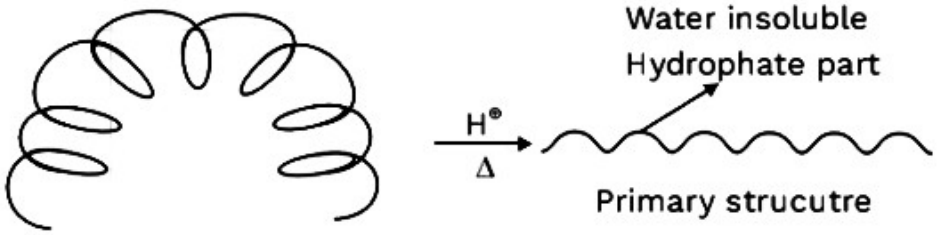




Intermolecular H-bonding

Tertiary Structure :





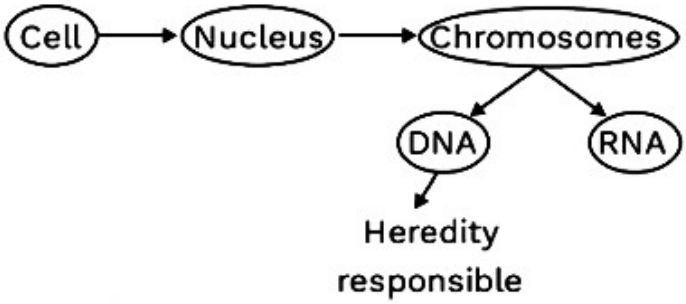
Quaternary Structure :

It's a combination of different 2° and 3° proteins.

Q7 **In de-naturation of protein which structure is entad?**

Sol After hydrolysis, Primary structures or polypetide chain and amino acids formed. Secondary and tertiray strucutres are destroyed.

DNA (Deoxyribonucleic Acid) And RNA(Ribonucleic Acid)



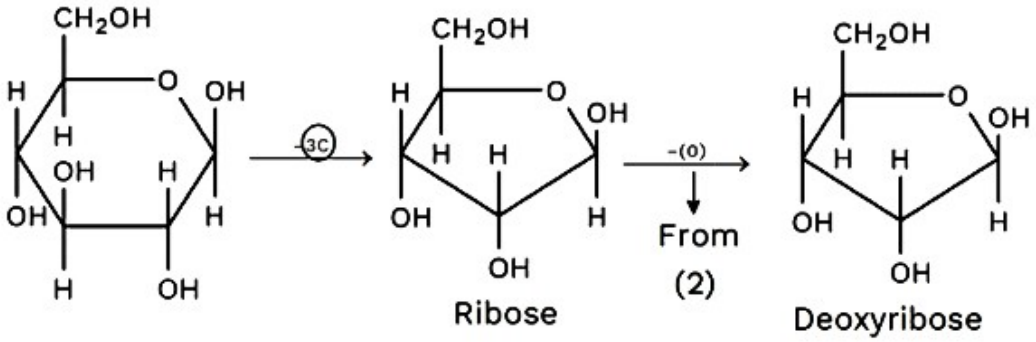
- **Components :**

1. Ribose Sugar
2. N-Base (Nitrogeneous)
3. H_3PO_4

→ Nucleoside

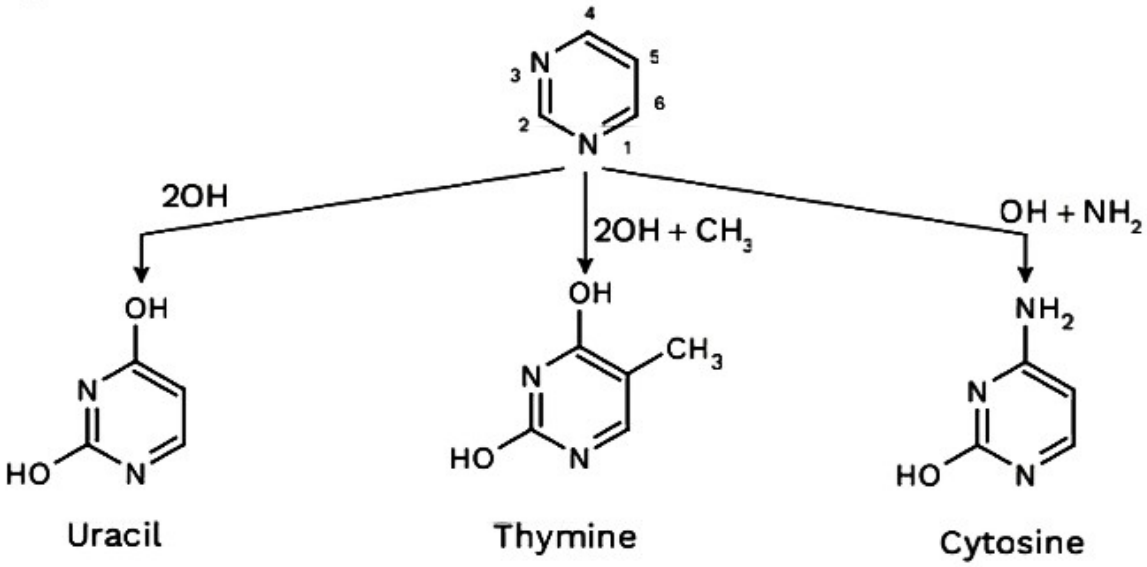
→ Nucleotide

- **Ribose Sugar :**

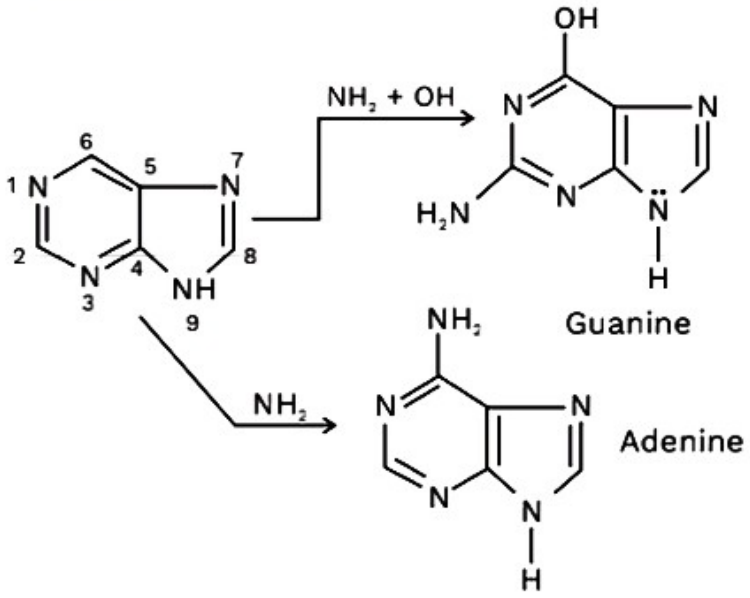


- **N-Base**

1. Pyrimidine



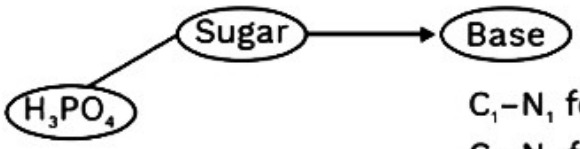
2. Pyrine :



DNA ⟶ CGAT

RNA ⟶ CGAU

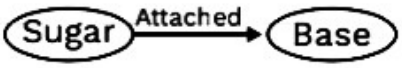
Nucleotide

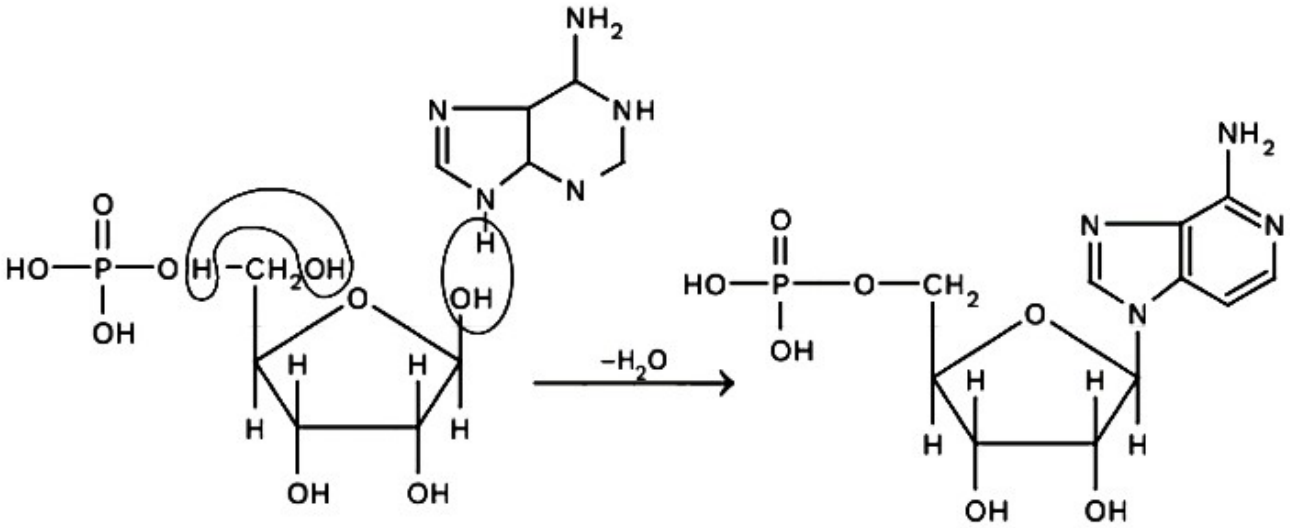


C_1–N_1 for pyrimidine

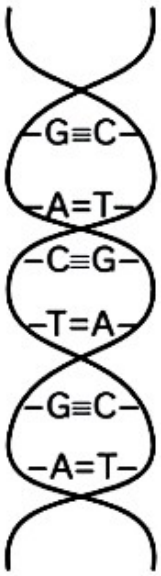
C_1–N_9 for pyrine

Nucleoside





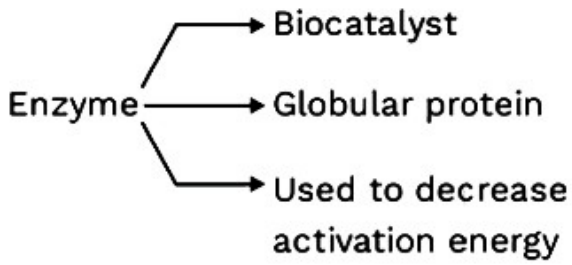
DNA	RNA
2-deoxy ribose	Ribose sugar
N-base C,G,A,(T)	N-Base C, G, A, (U)
Controlling	It synthesis protein
Double strained Helical Structure	Single strained Helical Structure





G−C → 3 H-bond
A−T → 2H-bond

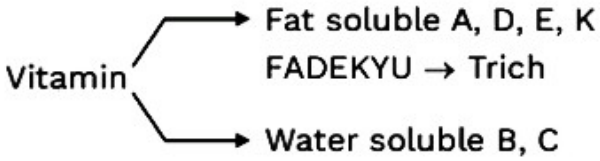
Enzyme :



Biuret-test and molisch test → Pending
histamine
Enzyme → 2° suffix :- abe

Maltose $\xrightarrow[H_3O^{\oplus}]{\text{Maltase (enzyme)}}$ glucose

Enzyme which oxidise one unit and reduce other unit are known as oxidoreductose.
Vitamin ⟶ vital + amine



A → Night Blindness
B_1 → Beri-Beri
B_2 → Cheilosis
B_6 → Conveusions
B_{12} → Anaemia
C → Scurvy
D → Rickets, Osteomalacia
E → Muscle weakness, fragility of RBC's
F → Increase blood clotting time

Test for protein or polypeptide (Biuret test)

$$\text{Urea} \xrightarrow{180^\circ C} \text{Biuret}$$

$$\text{Polypeptide or protein} + \text{Biueret} \xrightarrow[\text{or } CuSO_4]{\text{Cu(ii)iran}} \text{Violet colour complex (Coordination compound)}$$

(+ve) test

Q.8 **Glucose and Fructose are**

(A) Optical isomers **(B) Functional group isomers**

(C) Chain isomers **(D) Position isomers**

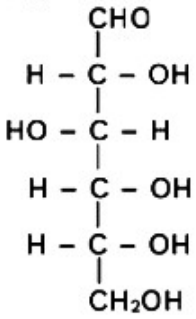
Sol. **[B]**

Glucose and fructose are aldhexose and ketohexose and both are functional isomers

Q.9 **The number of asymmetric carbon atom in the glucose molecule is**

(A) One **(B) Two** **(C) Four** **(D) Six**

Sol. **[C]**



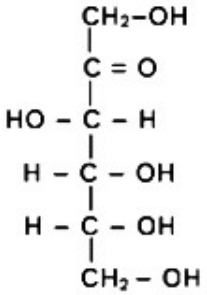
Number of chiral carbon (asymmetric) carbon atom is = 4

Q.10 **Which of the following is a ketohexose**

(A) Glucose **(B) Fructose** **(C) Sucrose** **(D) Starch**

Sol. **[B]**

Fructose is ketohexose



Q.11 **The change in optical rotation with time of freshly prepared solution of sugar is known as**

(A) Specific rotation **(B) Inversion**

(C) Rotation **(D) Mutarotation**

Sol. **[B]**

Q.12 **α-D glucose & β-D glucose are called**

(A) Anomer **(B) Enantiamer**

(C) Geometrical isomer **(D) All of these**

Sol. **[A]**

Anomer

Q.13 **Glucose is a/an**

(A) Oxidising sugar **(B) Reducing sugar**

(C) Reducing & oxidising sugar **(D) None of the above**

Sol. **[B]**

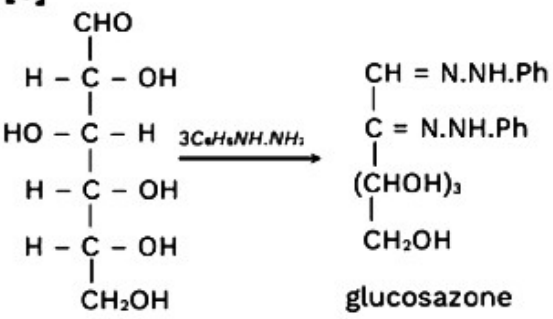
Glucose is reducing sugar because it give positive test with tollens reagent

Q.14 **Glucose with excess of phenyl hydrazine forms**

(A) Fructosazone **(B) Glucose phenyl hydrazone**

(C) Glucosazone **(D) Phenyl hydrazone of glucosazone**

Sol. **[C]**



glucosazone

Q.15 **Glucose when heated with CH_3OH in presence of dry HCl, α & β–methyl glucoside are formed. This is because it contain**

(A) An aldehyde group **(B) $-CH_2-OH$ group**

(C) A ring structure **(D) Fine hydroxyl group**

Sol. **[C]**

α and β methyl glycoside exist in ring structure

Q.16 **Alpha–glucose & beta–glucose are called**

(A) Isomers **(B) Anomers** **(C) Tautomers** **(D) Epimers**

Sol. **[B]**

α-D(+) & β-D(+) glucose are called anomers

Q.17 **A disaccharides which give only glucose on hydrolysis**

(A) Lactose **(B) Fructose** **(C) Sucrose** **(D) Maltose**

Sol. **[D]**

Maltose on hydrolysis give only glucose

Q.18 **When sucrose is heated with conc. HNO_3 the product is**

(A) Sucrosenitrate **(B) Oxalic acid**

(C) Formic acid **(D) Citric acid**

Sol. **[B]**

On heating with conc. HNO_3 sucrose gives oxalic acid

Q.19 **Hydrolysis of sucrose is called**

(A) Saponification **(B) Inversion**

(C) Mutarotation **(D) Rotatory motion**

Sol. **[B]**

Hydrolysis of sucrose gives glucose and fructose and it is called inversion

Q.20 **Starch can be used as an indicator for the detection of traces of**

(A) Proteins in blood **(B) Urea in blood**

(C) I_2 in aqueous solution **(D) Glucose in aqueous solution**

Sol. **[C]**

Q.21 $(C_6H_{12}O_5)_n + nH_2O \xrightarrow{(X)} C_{12}H_{22}O_{11} + H_2O \xrightarrow{(Y)} 2C_6H_{12}O_6$

Enzymes X & Y are

(A) Zymase & Invertase **(B) Diatase & maltase**

(C) Maltase & Diatase **(D) Diatase & zymase**

Sol. **[B]**

$(C_6H_{12}O_5)n + n\ H_2O \xrightarrow[diastase]{} C_{12}H_{22}O_{11} + H_2O \xrightarrow[H_2O]{Maltase} 2C_6H_{12}O_6$

Q.22 **The pH value of a solution in which a polar amino acid does not migrate under the influence of electric field is called**

(A) Iso-electric point **(B) Iso-electronic point**

(C) Neutrilization point **(D) None**

Sol. **[A]**

Q.23 The structure of amino at pH 6 is

(A) H_2N-CH_2COOH (B) $H_2N-CH_2-COO^-$

(C) $H_3N^+-CH_2COOH$ (D) $H_3N^+-CH_2COO^-$

Sol. [D]

Q.24 Individual amino acid can be distingused by their

(A) Action of HI (B) Carbylamine reaction

(C) Reduction (D) Isoelectric point

Sol. [D]

Q.25 $X + 3NH_3 \longrightarrow Y \xrightarrow{H^+/H_2O} H_2N-CH_2-COOH$, compound X is

(A) Chloroacetic acid (B) Bromoacetic acid

(C) Both A & B (D) Acetic acid

Sol. [C]

$$CH_2(Br)-CN \xrightarrow{NH_3} CH_2(NH_2)-CN \xrightarrow[\Delta]{H_2O/H^+} CH_2(NH_2)-COOH$$

Q.26 Which of the following is a fibrous protein

(A) Haemoglobin (B) Albumin (C) Keratin (D)Enzymes

Sol. [C]

Q.27 The sequence in which α–amino acid are linked to one another in a protein molecules is called

(A) Quaternary structure (B) Primary structure

(C) Tertiary structure (D) Secondary structure

Sol. [B]

Q.28 **The disease sickel–cell anaemia is caused by**

(A) Deficiency of protein **(B) Deficiency of vitamins**

(C) Defective haemoglobin **(D) Phenyl alanine hydroxylase**

Sol. **[C]**

Q.29 **Compare the reactions of an aldohexose and a 2-ketohexose with**

(i) Tollen's reagent,

(ii) Fehling's solution,

(iii) Benedict's reagent, and

(iv) Br_2/H_2O. Describe the changes that signal a positive test .

Sol. **(a)** These reagent all oxidize CHO to COOH or its salt in basic solution.

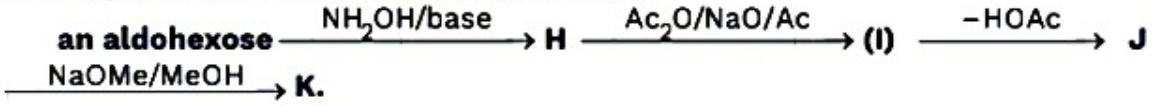
(i) Both give a positive test. Ketosugar reacts because it rearrange to an aldosugar under the basic conditions.

(ii) The formation of a red precipitate of Cu_2O denotes a positive test, given by both sugar.

(iii) Benedict's reagent is the Cu^{2+} citrate complex in NaOH and a positive test is indicated by the same changes as with Fehling's solution.

(iv) Only the aldosugar reacts a positive test is indicated by loss of the orange colour of the aqueous Br_2 solution.

Q.30 **(a) Supply structures for H through K given:**



(b) Explain the last step.

(c) What is the net structural change.

(d) Name this overall method,

(e) Discuss the possibility of epimer formation.

Sol. **(a)** E is an aldonic acid, F is its calcium salt,

$[HOCH_2CHOHCHOHCHOHCHOHCOO^-]_2$, Ca^{2+}, G is an aldopentose,

$HOCH_2CHOHCHOHCHOHCHO$.

(b) A sortening of the carbon chain by one C.

(c) This oxidative decarboxylation is called the *Ruff* degradation.

(d) The α–CHOH is oxidized to –CH = O without any configurational changes of the other chiral C's. No epimers are formed.

Q.31 **(a) Why do aldoses react with Fehling's solution and $PhNHNH_2$, but not with $NaHSO_3$.**
(b) Would you expect glycosides to react with either Fehling's or Tollen's reagent. Explain.

Sol. **(a)** These reaction are typical for the CHO group, which means that the open chain aldehyde form is in equilibrium with the cyclic hemiacetal form. Since Fehling's and osazone reactions are irreversible, the equilibrium shifts to restore the low concentrations (0.02%) of the aldehyde as it is depleted through reaction and eventually all the aldose reacts. Bisulphite addition is reversible and enough aldehyde remains in equilibrium with the bisulphite adduct to satisfy the equilibrium with the hemiacetal form. Consquently, there is no noticeable reaction.

(b) No, Glycosides, like simple acetal, are stable in these basic solutions. There is no –CH = O present to react. Glycosides are nonreducing; they do not reduce Cu^{2+} to Cu^{+} (as CuO).

Q.32 **The β– and α–D–glucose have different specific rotations. When either anomer is dissolved in water, their rotation change until the same fixed value results.**
(a) What term is used for these changes.
(b) Explain the change.

Sol. **(a)** This rotation change is called mutarotation.

(b) The α– and β–D–glucose are each in equilibrium with the open chain aldehyde form the therefore with each other.

$$\alpha\text{–D–Glucose} \rightleftharpoons \text{aldehyde form} \rightleftharpoons \beta\text{–D–Glucose}$$

As each anomer beings to establish this equilibrium, its specific rotation changes. When equilibrium is reached, the experimentally determined rotation remains constant. A base such as NaOH catalyzes the attainment of the equilibrium.

Q.33 **Explain the following**

(a) Why do glucose, mannose and fructose form identical osazone.

(b) Aldoses give positive Tollen's, Fehling's and osazone reactions but fail to respond Schiff's and bisulphite tests.

(c) The amino acid in water acts as an ampholyte.

(d) In an electric field, the amino acid migrates towards cathode when pH is below the isoelectric point while it migrate towards anode when pH is higher than isoelectric point.

Sol. **(a)** The formation of osazone involves C-1 and C-2 glucose, mannose and fructose have identical configurational at C-3, C-4 and C-5. Hence, they form same osazone.

(b) The open chain from is very very small and Tollen's, Fehling's and Osazone reactions are irreversible in nature, the equilibrium is thus shifted to restore this small concentration till whole of the aldoses react. Schiff's and bisulphite reactions are reversible, the equilibrium is thus not disturbed and open chain from having –CHO group is not available in good concentration.

(c) $\underset{\text{Cation}}{H_3\overset{+}{N}CHRCOOH} + OH^- \xrightleftharpoons{\text{Acting as an acid}} \underset{\text{Ampholy}}{H_3\overset{+}{N}CHRCOO^-} + H_2O$

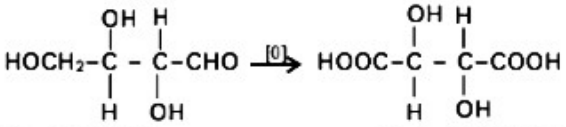
$$\downarrow \text{Acting as base}$$

$$\underset{\text{Anion}}{H_2NCHRCOO^-} + H_2O$$

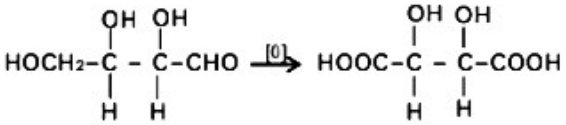
(d) Isoelectric point is the pH at which the amino acid has net zero charge and exists as dipolar ion ($H_3\overset{+}{N}CHRCOO^-$). When pH is below the isoelectric point, the cation ($H_3\overset{+}{N}CHRCOO^-$) predominates and it migrates to cathode while at pH higher than isoelectric point the anion ($H_2NCHRCOO^-$) predominate and migrate to anode.

Q.34 **Differentiate between D-erythose and D-threose using mild oxidation and reduction separately.**

Sol. Threose gives an optical active enantiomer while erythose gives the optically inactive meso isomer on mild oxidation

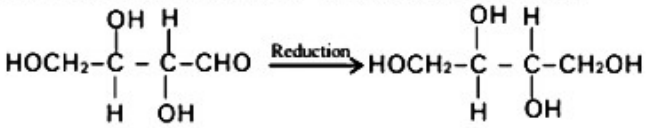


D– Threose D–(–) Tartaric acid

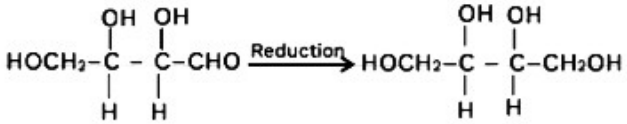


D– Erythrose Meso–tartaric acid

Thereose gives optical active enantiomer while erythose gives the optically inactive meso-isomer on mild reduction.



D–Threose D-2,2,3,4-butannetetrol



D–Erythrose Meso-1,2,3,4-butanetetrol

Q.35 **How the following conversions are made**

(i) Glucose into osazone, **(ii) Glucose into glucaric acid,**

(iii) Glucose into sorbitol, **(iv) Glucose into n–hexane,**

(vi) Fructose into glucose.

Sol. **(i)** Treat glucose with excess of phenylhydrazine.

(ii) Treat glucose with nitric acid.

(iii) Treat glucose with $NaBH_4$ or H_2/Pd.

(iv) Treat glucose with phosphorous and hydrogen iodide.

(v) Reduce fructose with H_2 in the presence of Ni and treat the alcohol with Fenton's reagent.

Q.36 **What happen when–**

(i) Glucose is heated with phenyl hydrazine,
(ii) Glucose is treated with hydroxylamine,
(iii) Cane sugar is warmed with dilute hydrochloric acid,
(iv) Glucose is made to react with Tollen's reagent,
(v) Glucose reacts with bromine water,
(vi) Glucose solution is heated with Fehling's solution,
(vii) Glucose is heated with conc. HNO_3,
(viii)Glucose is treated with HI in the presence of red phosphorous,
(ix) Alanine is treated with nitrous acid.

Sol. **(i)** $CH_2OH.(CHOH)_4CHO + H_2N.NHC_5H_6 \rightarrow CH_2OH(CHOH)_4CH=NNHC_6H_5$

Glucose phenyl hydrazone

With excess of phenyl hydrazine, glucosazone

$CH_2OH(CHOH)_3-\underset{\displaystyle CH=NNHC_6H}{\overset{\displaystyle |}{\underset{\displaystyle |}{C}}}=NNHC_6H_5$

is formed.

(ii) $CH_2OH(CHOH)_4CHO + H_2NOH \xrightarrow{¾} CH_2OH.(CHOH)_4CH=NOH$

glucose oxime

(iii) $C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{dil.HCl}} C_6H_{12}O_6 + C_6H_{12}O_6$

Cane sugar Glucose Fructose

(iv) $C_6H_{12}O_6 + 2Ag(NH_3)_2OH \rightarrow \underset{\displaystyle COOONH_4}{\overset{\displaystyle CH_2OH}{\overset{\displaystyle |}{\underset{\displaystyle |}{(CHOH)_4}}}} + 2Ag + 3NH_3 + H_2O$

Silver mirror

(v) $C_6H_{12}O_6 + [O] \xrightarrow{Br_2/H_2O} C_6H_{12}O_7$

Gluconic acid

(vi) $C_6H_{12}O_6 + 2Cu(OH)_2 + NaOH \rightarrow CH_2OH.(CHOH)_4COONa + Cu_2O + 3H_2O$ red ppt.

(vii) $CH_2OH(CHOH)_4CHO + 2[O] \rightarrow COOH.(CHOH)_4COOH$

(viii) n–Hexane is formed

(ix) $\underset{\displaystyle NH_2}{\overset{\displaystyle |}{CH_3CHCOOH}} + HONO \longrightarrow \underset{\displaystyle NH_2}{\overset{\displaystyle |}{CH_3CHCOOH}} + N_2 + H_2O$

Q.37 **Prove by giving one chemical reaction that**
(i) Glucose has carbonyl group,
(ii) Glucose has an aldehydic group,
(iii) Glucose has hydroxyl group,
(iv) Glucose is an aldose while fructose is a ketose,
(v) Cane sugar is a disaccharide

Sol. **(i)** Glucose reacts with HCN, NH_2OH and phenylhydrazine respectively.

(ii) Glucose can be oxidised to gluconic acid by bromine water, Tollen's solution of Fehling's solution.

(iii) Glucose undergoes acetylation with acetyl chloride.

(iv) Glucose can be oxidised with bromine water but fructose does not give this reaction.

(v) Cane sugar on hydrolysis gives one molecule of glucose and one molecule of fructose i.e., two monomers.

Q.38 **Discuss the following terms**
(a) thermoplastic
(b) thermosetting polymers
(c) elastomers
How are cross link introduced in natural rubber.

Sol. **(a) Thermoplastics:** These are the polymers that soften on heating and can be moulded into any shape. Intermolecular forces in them are intermediate between fibres and elastomers.

(b) Thermosetting polymer: These polymers are very hard, infusible and have extensive cross links. Example bakelite.

(c) Elastomers: In this type of polymers, the polymeric chains are held by weak intermolecular forces. Example : rubber

Heating natural rubber with sulphur causes cross linking of polymer chain through disulphide bond. This process is known as vulcanisation"

Q.39 **Which of the following pair, will give positive test with Tollen's reagent :**

[IIT-Scr. - 2004]

(A) Glucose, sucrose **(B) Sucrose, fructose**

(C) Glucose, fructose **(D) Acetophenone, hexanal**

Sol. **[C]**

Glucose (an aldehyde) and fructose (a hydroxyketone) both give positive test with tollen's reagent. Therefore option (C) is correct.

Q.40 **D-Glucopyranose exist in two forms** **[IIT- 2005]**

(A) Anomers **(B) Epimers**

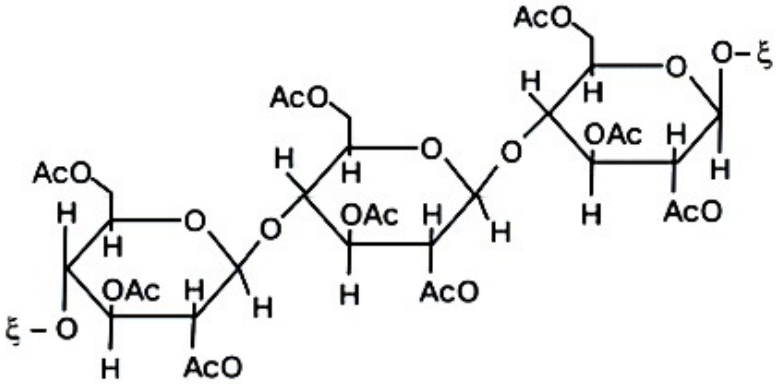
(C) Enantiomers **(D) Diastereoisomers**

Sol. **[A]**

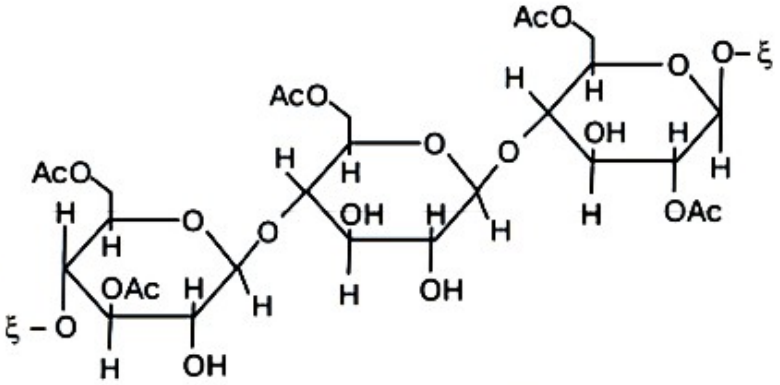
The two isomeric forms (α-and β-) of D-glucopyronose differ in configuration only at C-1: hence these are called anomers. Therefore option (A) is correct.

Q.41 **Cellulose upon acetylation with excess acetic anhydride/H_2SO_4 (catalytic) gives cellulose triacetate whose structure is –** **[IIT - 2008]**

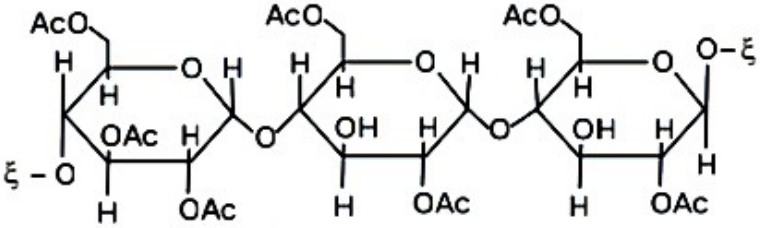
(A)



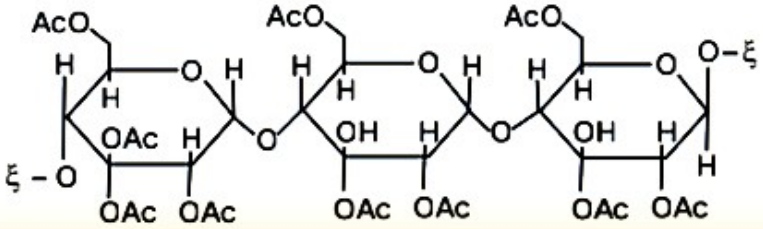
(B)



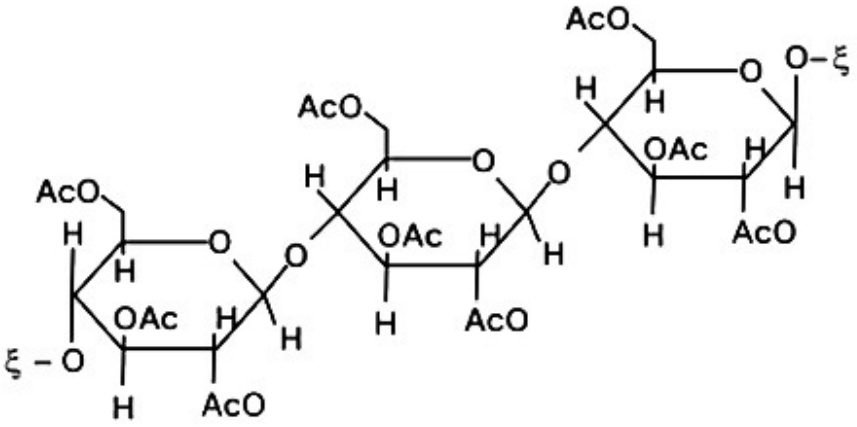
(C)



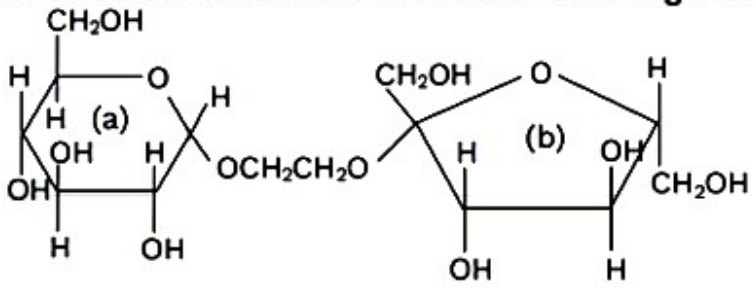
(D)



Sol. **[A]**



Q.42 **The correct statement about the following disaccharide is** **[IIT - 2010]**

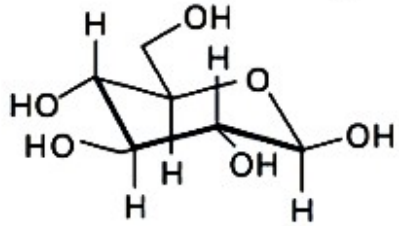


(A) Ring (a) is pyranose with α-glycosidic link
(B) Ring (a) is furanose with α-glycosidic link
(C) Ring (b) is furanose with α-glycosidic link
(D) Ring (b) is Pyranose with β-glycosidic link

Sol. **[A]**

Ring (a) is pyranose with α-glycosidic link

Q.43 **The following carbohydrate is** **[IIT - 2011]**

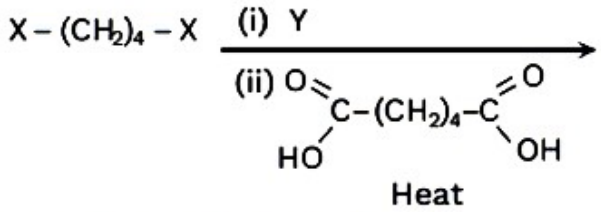


(A) a ketohexose **(B) an aldohexose**
(C) an α-furanose **(D) an α-pyranose**

Sol. **[B]**

Aldohexose

Q.44 **The correct functional group X and the reagent / reaction conditions Y in the following scheme are** **[IIT - 2011]**



(A) X = $COOCH_3$, Y = H_2/Ni/Heat **(B) X = $CONH_2$, Y = H_2 / Ni / heat**

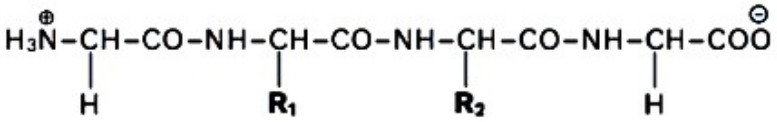
(C) X = $CONH_2$, Y = Br_2/ NaOH **(D) X = CN, Y = H_2 / Ni / heat**

Sol. **[C, D]**

Factual

The most appropriate answer to this question is (A,B,C,D). But because of ambiguity in language, IIT has declared (C & D) as correct answer

Q.45 **The substituents R_1 and R_2 for nine peptides are listed in the table given below. How many of these peptides are positively charged at pH = 7.0 ?** **[IIT-2012]**



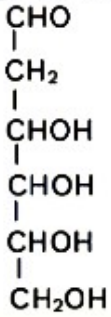
Peptide	R_1	R_2
I	H	H
II	H	CH_3
III	CH_2COOH	H
IV	CH_2CONH_2	$(CH_2)_4NH_2$
V	CH_2CONH_2	CH_2CONH_2
VI	$(CH_2)_4NH_2$	$(CH_2)_4NH_2$
VII	CH_2 COOH	CH_2CONH_2
VIII	CH_2 OH	$(CH_2)_4NH_2$
IX	$(CH_2)_4$ NH_2	CH_3

Sol. **[D]**

IV, VI, VIII, IX

Four amides are positively charged at pH = 7.0

Q.46 **When the following aldohexose exists in its D-configuration, the total number of stereoisomers in its pyranose form is** **[IIT - 2012]**



Sol. **[8]**

8 stereoisomers possible for given compound in pyranose form of D-configuration